INCH-POUND
MIL-H-2815G
17 February 1989
SUPERSEDING
MIL-H-2815F
5 June 1978

MILITARY SPECIFICATION

HOSE ASSEMBLIES, RUBBER, DIVER'S BREATHING AIR AND GAS SUPPLY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for rubber hose assemblies for underwater use in supplying air or other breathing gas mixtures to divers at water depths to 600 feet using surface supplied diving outfits.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

FEDERAL

QQ-B-637 -Brass, Naval: Rod, Wire, Shapes, Forgings, and Flat Products with Finished Edges (Bar, Flat Wire, and Strip).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4220

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T15:44Z
Check the source to verify that this is the current version before use.

PPP-F-320 - Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.

MILITARY

MIL-H-775 - Hose, Hose Assemblies; Rubber, Plastic, Fabric, or Metal (Including Tubing), and Associated Hardware: Packaging of
MIL-R-3065 - Rubber, Fabricated Products.
MIL-S-5059 - Steel, Corrosion-Resistant (18-8), Plate Sheet and Strip
MIL-L-19140 - Lumber and Plywood, Fire-Retardant Treated

STANDARDS

FEDERAL

FED-STD-162 - Hose, Rubber, Visual Inspection Guide for.
FED-STD-601 - Rubber: Sampling and Testing.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Non-Government publications. The following document forms a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 573 - Standard Test Method for Rubber-Deterioration in an Air Oven. (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets or MS standards), the text of this document takes precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T15:44Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 First article. When specified (see 6.4), a sample shall be subjected to first article inspection (see 6.4) in accordance with 4.3.

3.2 Materials. Material shall be as specified herein. Material not definitely specified shall be of the quality regularly used by the contractor in these types of hose assemblies. Material shall be new and free from all defects and imperfections that may adversely affect the serviceability or appearance of the finished product. When specified a certificate of compliance shall be prepared (see 6.3).

3.2.1 Inner tube. The basic rubber stock of which the inner tube is fabricated shall be natural rubber or a synthetic rubber. Reclaimed rubber or rubber containing pine tar or coal tar shall not be used.

3.2.2 Cover. The basic rubber stock of which the cover is fabricated shall be polymerized chloroprene.

3.2.3 Fabric reinforcement. Fabric reinforcement shall consist of two or more plies of braided cotton, rayon, or polyester (see 3.3.2).

3.2.4 Couplings and clamps. The couplings and nut for female coupling shall be made from brass conforming to copper alloy number 464 specified in QQ-B-637 and the clamps from stainless steel conforming to MIL-S-5059, type 316. Washer material shall be in accordance with MIL-R-3065.

3.2.5 Organic solvents. The organic solvents used at any stage in the construction of the diver's hose, including the solvents used for rubber compounds for fastening couplings to the hose, for sealing or capping the ends of the hose, for bonding the layers and plies of the hose together, or for any other application that might contribute to the contamination of an air or gas mixture stream passing through the hose, shall conform to the following requirements.

3.2.5.1 Composition. When specified a certificate of compliance shall be prepared (see 6.3).

3.2.6 Other rubber additives. The hose shall contain the minimum quantities necessary of only the purest grades of rubber accelerators, both in the cover and inner tube in order to be as free from odor as possible. When specified a certificate of compliance shall be prepared (see 6.3).

3.2.7 Coupling lubricants and washer dressing. No coupling lubricants or washer dressings, except water, shall be used.

3.2.8 Recovered materials. Unless otherwise specified herein, all material and articles incorporated in the products covered by this specification shall be new and may be fabricated using materials produced from

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T15:44Z
Check the source to verify that this is the current version before use.

recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3.3 Construction. The hose assembly shall consist of an inner rubber tube, two or more plies of fabric braid reinforcement (see 3.2.3) with a rubber ply between the braids, a cover, couplings and clamps. Each end of the hose shall be capped or sealed with a rubber compound 1/16-inch thick of the same composition as the cover; the cap or seal shall prevent the ingress of water into the braid.

3.3.1 Tube and cover. The inner tube shall be homogeneous, free from extraneous inclusions, nonporous, water resistant, and of uniform thickness. The tube shall be smooth and free of adhering soapstone dust or other particles. The tube and cover shall be free from pitting and free from substances such as dirt, grease, or metal shavings that may affect the material characteristics or appearance. The cover may have spiral markings from a tightly wrapped cure and may have pinpricking when specified (see 6.2).

3.3.2 Fabric reinforcement. The fabric reinforcement shall consist of two or more plies of braided cotton, rayon or polyester, well embedded in a rubber compound with a rubber ply between the braids. The reinforcement shall be evenly and uniformly braided, free from dirt, knots, lumps, irregularities of twist or other defects.

3.3.3 Couplings. Each 100-foot and 3-foot length of hose shall have one male and one female coupling fitted into alternate ends of the hose. The couplings shall conform to the dimensions and characteristics shown on figures 1 and 2. Each coupling shall be secured in place by three metal clamps (see 3.3.4) and cement. Couplings shall be interchangeable.

3.3.4 Clamps. Each coupling shall be secured on the hose end by means of three metal clamps staggered at 120° degrees. The clamps shall conform to the dimensions and characteristics of those shown on figure 3. Clamps shall be tightened until the clamp depresses the hose so that daylight appears when a straightedge is held along the outside surface of the hose and is at 180° degrees to the buckle.

3.4 Length. The lengths of hose shall be as follows, as specified (see 6.2):

(a) Three feet, plus or minus 1 inch, including couplings.
(b) One-hundred feet, plus or minus 1 foot, including couplings.

3.5 Physical requirements. Each hose shall conform to the following physical requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T15:44Z
Check the source to verify that this is the current version before use.

Size:	
side diameter (inch)	1/2 ± 1/32
Outside diameter (inch)	1 ± 1/32
Thickness of inner tube (minimum) (inch)	0.060
Thickss of cover (minimum) (inch)	0.060
Adhesion:	
Between tube and plies of braid (minimum) (pounds)	10
Between plies of braid (minimum) (pounds)	12
Between plies of braid and cover (minimum) (pounds)	10
Initial tensile strength (minimum) (pounds-force per square inch)	
Tube	1200
Cover	1800
Elongation (minimum) (percent)	
Tube	200
Cover	300

3.6 After oven aging. The tensile strength of the tube and cover shall show a change from the initial tensile strength of not more than 25 percent (see 4.6.3).

3.7 Immersion. The tensile strength of the cover shall show a decrease from the initial tensile strength of not more than 50 percent (see 4.6.4).

3.8 Change in volume. The change in volume of the cover stock in the special testing fuel shall be not greater than 50 percent (see 4.6.5).

3.9 Hydrostatic pressure. Hydrostatic pressure shall be as specified in 3.9.1 and 3.9.2.

3.9.1 Bursting strength. The hose shall be constructed so it will not burst at any pressure less than a gauge pressure of 2,400 pounds per square inch (lb/in^2) (see 4.6.6).

3.9.2 Proof strength. The hose shall be constructed to withstand a gauge pressure of 1200 lb/in^2 for 1 minute without exposing imperfections (see 4.6.7).

3.10 Elongation or contraction. The hose shall be constructed so that the length change shall be less than plus 2 or minus 4 percent at a gauge pressure of 600 lb/in^2 (see 4.6.8).

3.11 Coupling pull-out strength. The hose and couplings shall be so mated that they will withstand a tension load of 1,100 pounds for 1 minute with an internal gauge pressure of 600 lb/in^2 (see 4.6.9).

3.12 Physiological and off-gassing. The physiological effect and off gassing shall be as specified in 3.12.1 and 3.12.2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T15:44Z
Check the source to verify that this is the current version before use.

3.12.1 Physiological effect. The hose shall not contaminate the air or other breathing mixture through the hose with obnoxious or harmful vapors, gases, or particles (see 4.6.10 and 4.6.11).

3.12.2 Off-gassing. Hoses shall not contaminate the gas by amounts greater than those specified in table I.

TABLE I. Maximum allowable concentration of hydrocarbons in air passing through hose.

Hose length (feet)	Hydrocarbon concentration as methane equivalents mg/m^3
3	4
100	100

3.13 Branding. At the option of the manufacturer, the hose may be branded by either of the following methods.

(a) Each length of hose shall be branded with the manufacturer's name and trademark, the month and year of manufacture and "Diver's H.P." in the cover and vulcanized thereto. Hose 100 feet in length shall have brands inlaid in the cover at least two places on each length, approximately 4 feet from the ends. Letters shall be at least 1/4-inch high. Hose 3 feet in length shall have one inlaid brand approximately in the center.

(b) Each length of hose shall be branded by continuously embossing or inlaying in the cover the manufacturer's name and trademark, the month and year of manufacture, and "Diver's H.P.". The letters shall be at least 1/4-inch high.

3.14 Hose date. Hose dated more than 6 months before the date submitted for inspection shall not be accepted.

3.15 Workmanship. The tube and cover shall be free of pitting, dirt, grease, and metal shavings. The fabric reinforcement shall be uniformly braided, free of dirt, knots, lumps or irregularities of twist.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T15:44Z
Check the source to verify that this is the current version before use.

performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall consist of the examination and tests specified in table II (see 6.4). When specified (see 6.2), first article inspection shall be conducted at a certified testing laboratory. Two 100-foot and four 3-foot sample units shall be provided. When specified, a test report shall be prepared (see 6.3).

TABLE II. First article inspection.

Examination and test	Requirement	Test paragraph
Visual and dimensional examination	FED-STD-162	4.5.1
Off gassing	3.2.5, 3.12.2	4.6.10
Tensile strength and elongation	3.5	4.6.1
Change in volume	3.8	4.6.5
Adhesion	3.5	4.6.2
Oven aging	3.6	4.6.3
Immersion	3.7	4.6.4
Hydrostatic bursting strength	3.9.1	4.6.6
Hydrostatic proof	3.9.2	4.6.7
Longitudinal stability	3.10	4.6.8
Coupling pull-out strength (3-foot lengths only)	3.11	4.6.9

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T15:44Z
Check the source to verify that this is the current version before use.

4.4 Quality conformance inspection. Quality conformance inspection shall consist of the visual and dimensional examinations in 4.5 and the tests in 4.6. When specified test reports shall be prepared (see 6.3).

4.4.1 Lots. For purposes of sampling, a lot shall consist of all hose of the same length made at the same factory, of the same materials, of the same construction, and offered for inspection at the same time.

4.4.2 Sampling for visual and dimensional examination and nondestructive testing. Sampling for visual and dimensional examination and nondestructive testing shall be in accordance with table III.

TABLE III. Visual and dimensional examination.

Lot size	Sample size	Accept	Reject	1/ 2/ 3/
		Major defects		
2-150	13	0	1	
151-280	32	1	2	
281-500	50	2	3	
501-1200	80	3	4	
1201-3200	125	5	6	
3201-10000	200	10	11	
10001-35000	315	14	15	
35001-and over	500	21	22	
		Minor defects		
2-15	3	0	1	
16-50	8	1	2	
51-90	13	1	2	
91-150	20	2	3	
151-280	32	3	4	
281-500	50	5	6	
501-1200	80	10	11	
1201-3200	125	10	11	
3201 and over	200	14	15	

1/ All defective items must be replaced with acceptable items prior to lot acceptance.
2/ Inspect sample size until reject criteria is reached.
3/ Rejected lots may be secreened and resubmitted for inpsection and retest.

4.4.3 Sampling for destructive inspection. A random sample of hose shall be selected from each lot of 3-foot hose, or 3-foot lengths shall be fabricated by the manufacturer from each lot of 100-foot lengths of hose for the test specified in 4.5.3. The number of samples shall be as shown in table

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T15:44Z
Check the source to verify that this is the current version before use.

IV. The samples shall be capped or sealed with 1/16 inch of the tubing rubber compound and equipped with couplings in the same manner as completed hose.

TABLE IV. Sampling for destructive inspection.

Number of linear feet of hose in lot	Number of 3-foot hose lengths in sample
9,000 and under	6
9,001 to 15,000	7
15,001 to 25,000	10
25,001 to 40,000	15
40,001 to 60,000	20
60,001 to 100,000	30
100,001 to 150,000	40
150,001 and over	50

4.5 Nondestructive and destructive inspection. Nondestructive and destructive inspection shall be as specified in 4.5.1 through 4.5.5.

4.5.1 Visual and dimensional examination. Visual and dimensional examinations shall be in accordance with class A as specified in FED-STD-162 and table III herein.

4.5.1.1 Destructive. The examinations specified in FED-STD-162 that destroy or alter the sample shall only be made on the 3-foot lengths selected in accordance with 4.4.3.

4.5.1.2 Nondestructive. Nondestructive examinations shall be made on all lengths selected in accordance with 4.4.2.

4.5.2 Nondestructive inspection. Each sample length of hose selected in accordance with 4.4.2 shall be subjected to the visual and dimensional examination specified in 4.5.1.2 and the off-gassing test specified in 4.6.10. After successfully passing this examination and test, the hose shall be subjected to the remaining tests shown in table V.

TABLE V. Nondestructive inspection.

Examination and test	Requirement	Test
Hydrostatic proof	3.9.2	4.6.7
Longitudinal stability	3.10	4.6.8

4.5.3 Destructive inspection. If the lot has not been rejected as a result of the nondestructive tests specified in 4.5.2, the 3-foot lengths selected or fabricated in accordance with 4.4.3 shall be subjected to the destructive tests shown in table VI.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T15:44Z
Check the source to verify that this is the current version before use.

TABLE VI. Destructive inspection.

Examination and test	Requirement	Test
Tensile strength and elongation	3.5	4.6.1
Adhesion	3.5	4.6.2
Oven aging	3.6	4.6.3
Immersion	3.7	4.6.4
Change in volume	3.8	4.6.5
Hydrostatic bursting strength	3.9.1	4.6.6
Coupling pull-out strength (3-foot lengths only)	3.11	4.6.9

4.5.4 Action in case of failure. Acceptance of the first lot of hose from one manufacturer shall be withheld pending receipt of the test report on the quality conformance inspection samples representing the lot. At the option of the contractor, acceptance shall not be withheld pending receipt of the test report on the quality conformance inspection test samples representing a lot if the contractor has previously furnished acceptable hoses to the Government. However, when this is the case and the test report later reveals that the lot was unsatisfactory, four additional sample 3-foot sections shall be selected from every lot offered for delivery. The sample sections so selected shall be subjected to the test or tests wherein failure was observed. Lots shall then be accepted only upon receipt of a satisfactory test report on the samples so selected and shall be rejected if the lot is found to be unsatisfactory. This additional testing shall be discontinued and sampling procedures returned to the normal basis as specified in 4.4.2 when four successive lots have been accepted.

4.5.5 Disposition of samples. Samples subjected to the destructive tests shall not be submitted with the contract. Samples subjected to the nondestructive tests may be submitted with the contract.

4.6 Test procedures. Test procedures shall be as specified in 4.6.1 through 4.6.11.

4.6.1 Tensile strength and elongation. Tensile strength and elongation shall be conducted as specified in methods 4111 and 4121 of FED-STD-601 using die III, to determine conformance to 3.5.

4.6.2 Adhesion. Adhesion tests shall be conducted in accordance with method 8011 of FED-STD-601, to determine conformance to 3.5.

4.6.3 Oven aging. The hose shall be subjected to the oven aging test of ASTM D 573. The aging period shall be 70 hours at 100 degrees Celsius (^{0}C) to determine conformance to 3.6.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T15:44Z
Check the source to verify that this is the current version before use.

4.6.4 Immersion. The immersion test shall be conducted in accordance with method 6111 of FED-STD-601, using distilled water, except that the immersion period shall be 166 ± 1/4 hours at 70 ± 2°C to determine conformance to 3.7.

4.6.5 Change in volume in distilled water. The change in volume test shall be conducted in accordance with method 6211 specified in FED-STD-601 using distilled water, to determine conformance to 3.8, except that the immersion period shall be 166 ± 1/4 hours at 70 ± 2°C.

4.6.6 Hydrostatic bursting strength. The hose shall be subjected to a gauge pressure of 2400 lb/in^2 in accordance with method 10011 of FED-STD-601, to determine conformance to 3.9.1.

4.6.7 Hydrostatic proof pressure. The hose, complete with couplings, shall be subjected to a gauge pressure of 1200 lb/in^2 for 1 minute in accordance with method 10211 of FED-STD-601, to determine conformance to 3.9.2.

4.6.8 Longitudinal stability. The hose shall be subjected to a hydrostatic gauge pressure of 600 lb/in^2 in accordance with method 10311 of FED-STD-601, to determine conformance to 3.10.

4.6.9 Coupling pull-out strength. The 3-foot lengths, including couplings shall be held in the tensile device by clamping onto the hose couplings. The sample with a constant 600 lb/in^2 internal hydrostatic gauge pressure shall be loaded at a rate of 200 pounds of tension per minute until 1100 pounds is reached. The 1100-pound tensile load shall be maintained for 1 minute to determine conformance to the requirements specified in 3.11. Following this test, the hose shall show no signs of imperfections.

4.6.10 Off-gassing tests. Measurements shall be made of the off-gassing of organic compounds from hoses. A Beckman Total Hydrocarbon Analyzer, Model 400 or any equivalent process analyzer employing a hydrogen flame ionization detector having a range from 0 to at least 1,000 milligrams per cubic meter (mg/m^3) methane equivalents (see 4.6.10.1) shall be used. The measurements shall be made only on hoses that have not been flushed with air, gas, or water or that have been stored at a temperature no lower than 23°C for a period of at least 7 days since the hoses were flushed. During the 7-day storage period, the ends of the hoses shall either be sealed shut or fastened together so that the confined gas may not escape. Both the analyzer and hoses shall be maintained at a temperature not lower than 23°C throughout the testing period. The procedure for measuring off-gassing rates from hoses is specified hereinafter as the flush test (see 4.6.10.2).

4.6.10.1 Terminology. The hydrogen flame ionization detector of the Total Hydrocarbon Analyzer is sensitive not only to hydrocarbons themselves, but also to related compounds that will ionize in a hydrogen flame. In 4.6.10 through 4.6.10.2, the term hydrocarbon is intended to designate all organic compounds detected by the Total Hydrocarbon Analyzer. Likewise, the term

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T15:44Z
Check the source to verify that this is the current version before use.

methane equivalents designates the concentration of methane that will cause the instrument to give the same reading as the air or other gas tested. Each unit is expressed in terms of measurements made at 23^0C and at a pressure of 760 millimeters of mercury.

4.6.10.2 Flush test. By this procedure, measurements shall be made of the increase in the hydrocarbon concentration of a stream of air flowing through the hose at a velocity of 28 liters per minute (1 cubic foot per minute). The temperature of the hose, air supply, and analyzer shall be not less than 23^0C. A diagram of the flow arrangement is shown on figure 4. Before the air passes through the hose, the air shall be clean and shall contain not more than 1 mg/m^3 of hydrocarbons (methane equivalents). The analyzer shall be zeroed with air passing at the stipulated velocity and temperature through the connector tubes only. The hose shall then be inserted in the line and the airstream passed through it. For the ensuing 15 minutes, readings of the hydrocarbon concentration shall be recorded frequently, if not continuously. The hoses shall be rated on the reading at the end of the 15 minute test period. Hoses which contaminate the air by greater amounts than specified in table I are not acceptable (see 3.12.2).

4.6.11 Special physiological test. Contractor may be requested to arrange for special physiological test to be performed (see 6.2). Government agent requesting the test will stipulate the nature and details of the test at the time the test is requested.

4.7 Inspection of packaging. Sample packs, and the inspection of the preservation packing and marking for shipment stowage and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6.6)

5.1 Packaging requirements. Hose assemblies shall be preserved level A or C, packed level A, B or C as specified (see 6.2) and marked in accordance with MIL-H-775 and shall include bar codes and applicable packaging acquisition options therein as specified (see 6.2) in addition, for Navy acquisitions, the following applies:

(a) Navy fire-retardant requirements

(1) Lumber and plywood. When specified (see 6.2), all lumber and plywood including laminated veneer material used in shipping container construction members, blocking, bracing, and reinforcing shall be fire-retardant treated material conforming to MIL-L-19140 as follows:

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T15:44Z
Check the source to verify that this is the current version before use.

Levels A and B - Type II - weather resistant.
Category 1 - general use.

Level C - Type I - non-weather resistant.
Category 1 - general use.

(2) Fiberboard. When specified (see 6.2), fiberboard used in the construction of class-domestic, non-weather resistant fiberboard and cheated fiberboard boxes including interior packaging forms shall meet the flamespread and the specific optic density requirements of PPP-F-320.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The hose assemblies are intend to be used in diver's operations in water depths to 600 feet. The material handled may be fresh air or other breathing gas mixtures.

6.2 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number, and date of this specification.
(b) Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).
(c) When first article testing is required (see 3.1).
(d) If pinpricking on cover is acceptable (see 3.3.1).
(e) Lengths of hose required (see 3.4).
(f) Where first article inspection shall be conducted (see 4.3 and 6.4).
(g) Special physiological test if required (see 4.6.11).
(h) Level of packaging and marking required (see 5.1).
(i) When preservation and when fire retardant treatments or materials are required (see 5.1(a)).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T15:44Z
Check the source to verify that this is the current version before use.

Reference Paragraph	DID Title	DID Number	Suggested Tailoring
3.2, 3.2.5.1, 3.2.6	Certificate of compliance	DI-E-2121	
4.3	First article inspection report	DI-T-4902	
4.4	Inspection and test report	DI-T-5329	

The above DID's were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD 1423.

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the items(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, a standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested as specified in 4.3. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.5 Provisioning. Provisioning Technical Documentation (PTD), spare parts, and repair parts should be furnished as specified in the contract. When ordering spare parts or repair parts for the equipment covered by this specification, the contract should state that such spare parts and repair parts should meet the same requirements and quality assurance provisions as the parts used in the manufacture of the equipment. Packaging for such parts should also specified.

6.6 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.7. Destructive inspection. If any hose fails one or more of the tests (see 4.5.3), the entire lot represented by the sample will be rejected.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T15:44Z
Check the source to verify that this is the current version before use.

6.8 Subject term (key word) listing.

Clamps and couplings
Elongation
Equipment, diving
Hydrostatic pressure
Inner tube
Off-gassing
Oven aging
Underwater diving equipment

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - EA
Navy - SH

Review activities:
Army - MD
DLA - CS

User activities:
Army - ME
Navy - MS

Preparing activity:
Navy - SH
(Project 4220-0338)

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T15:44Z
Check the source to verify that this is the current version before use.

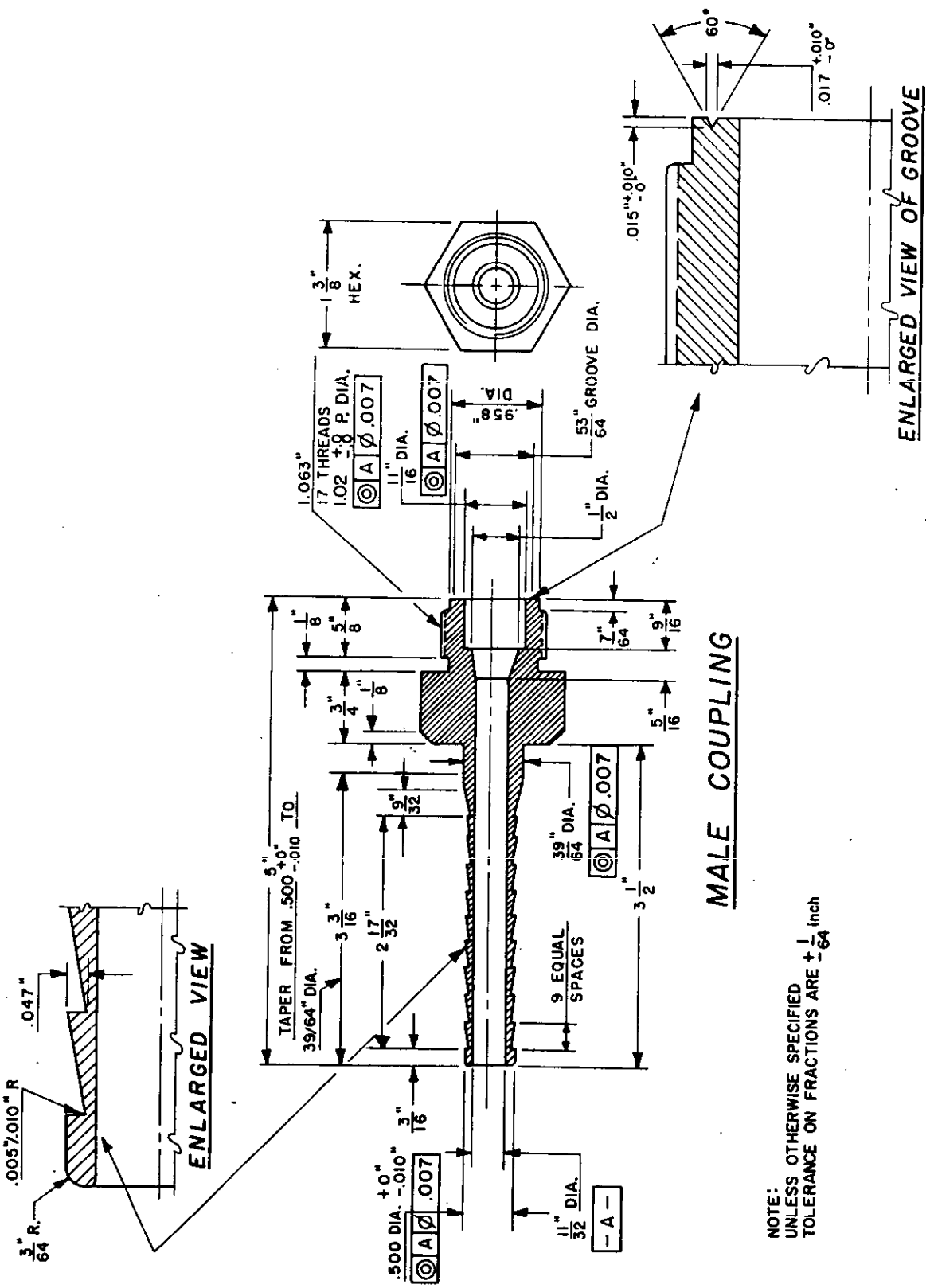
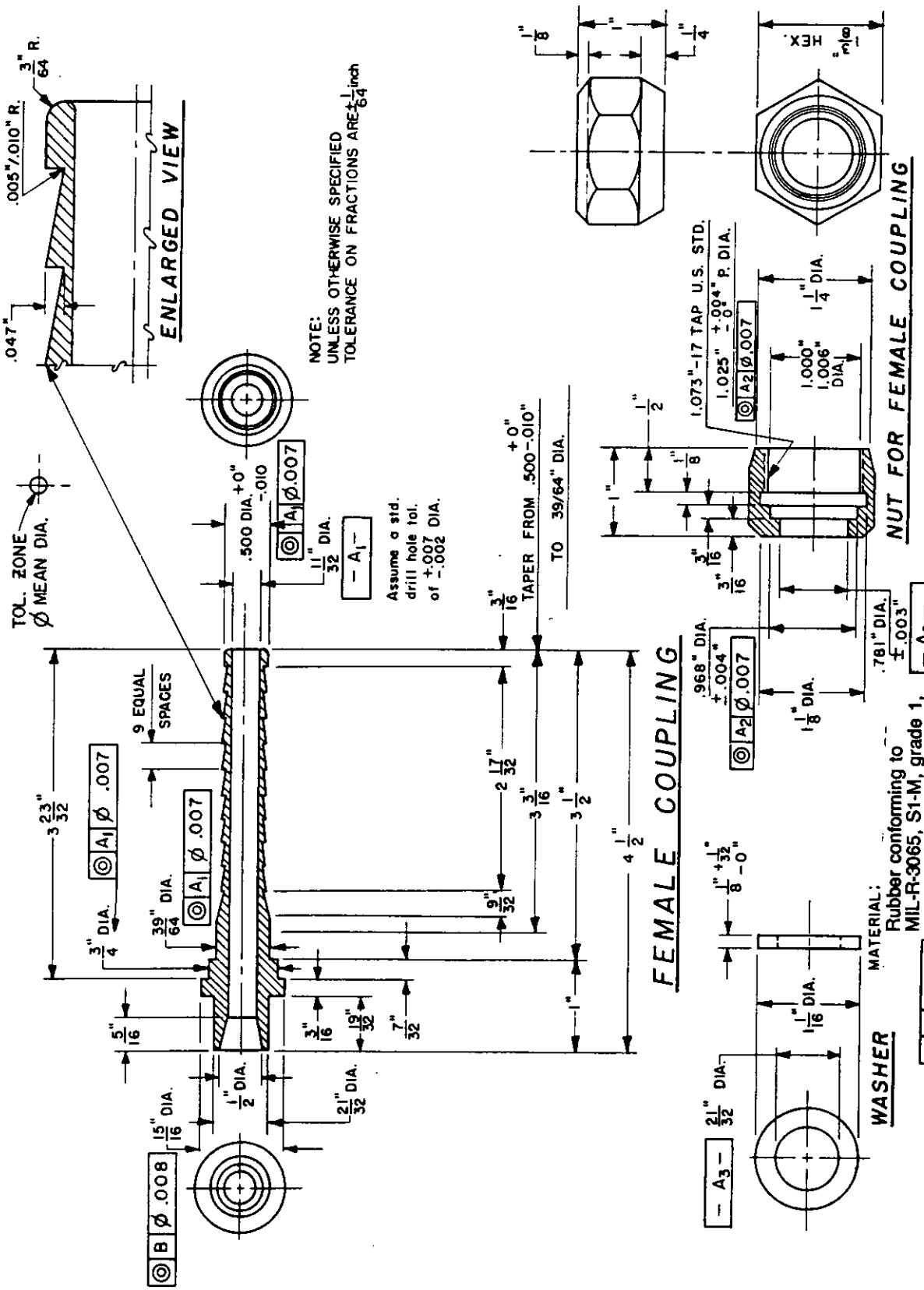


FIGURE 1. Male coupling.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T15:44Z
Check the source to verify that this is the current version before use.



SH 10133

FIGURE 2. Female coupling.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T15:44Z
Check the source to verify that this is the current version before use.

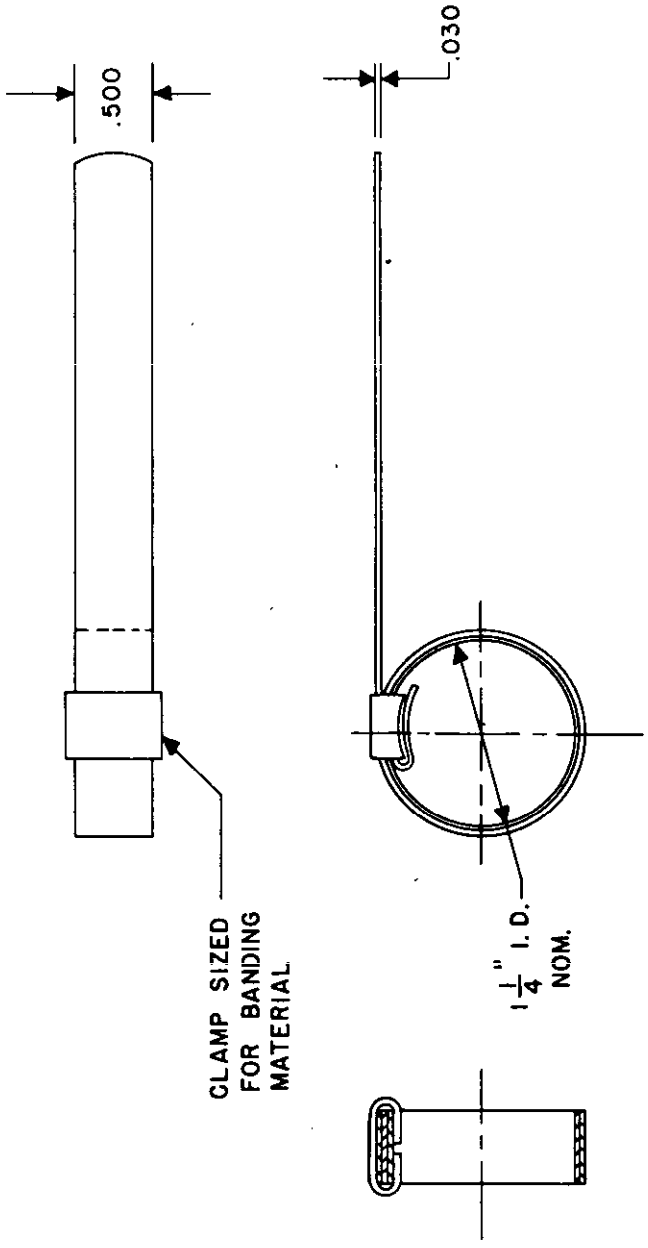
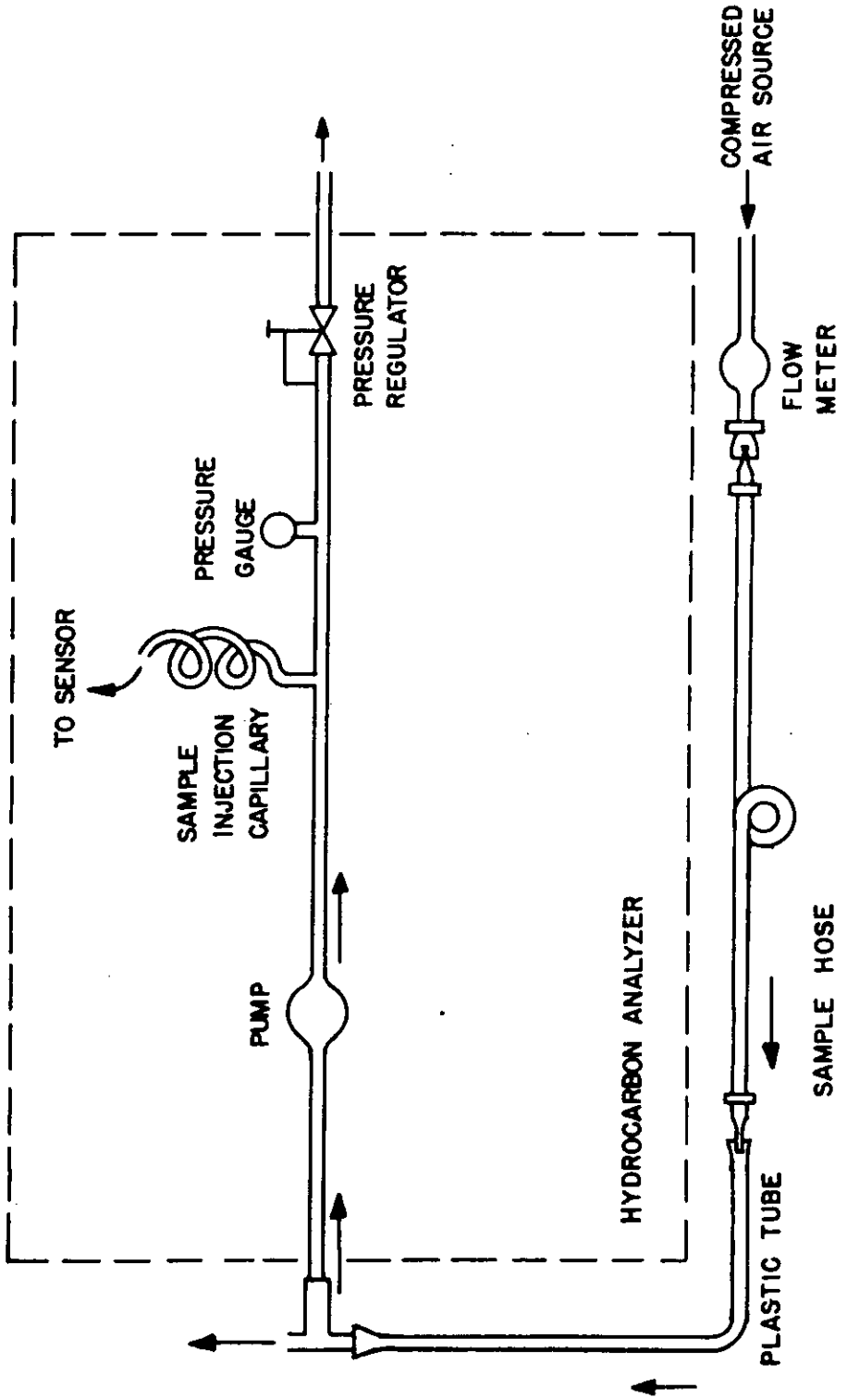


FIGURE 3. Clamp.

SH 10134

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T15:44Z
Check the source to verify that this is the current version before use.



SH 10136

FIGURE 4. Flow diagram of apparatus for measuring the concentration of hydrocarbons in a stream of air or other gas after it has passed through a test hose.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T15:44Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE NAVY

COMMANDER
NAVAL SEA SYSTEMS COMMAND (SEA 55Z3)
DEPARTMENT OF THE NAVY
WASHINGTON, DC 20362

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12503 WASHINGTON D. C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

COMMANDER
NAVAL SEA SYSTEMS COMMAND (SEA 55Z3)
DEPARTMENT OF THE NAVY
WASHINGTON, DC 20362

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T15:44Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-H-2815G	Hose Assemblies, Rubber, Diver's Breathing Air and Gas Supply

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T15:44Z
Check the source to verify that this is the current version before use.